Global Superstore

Retail Sales and Data Quality Assessment

Business Objective

Analyze historical sales transactions to identify trends, evaluate profitability, assess operational performance, and validate data quality to support executive decision-making.

Stakeholders

- CEO
- Sales Managers
- Operational Managers
- Finance Team
- Regional Managers

Business Questions

1. Which products generate the highest revenue?
2. Which product generates the highest profit?
3. Which products underperform in terms of profitability?
4. Which customer segment is most profitable?
5. Are shipping costs associated with differences in product profitability?
6. Which products should be considered for further profitability review?
7. Are discounts associated with lower profitability?
8. Are there any data quality issues that could affect reporting?

Key KPIs

KPI	Definition/Purpose	Status
Total Sales	Total revenue recorded across transactions	Calculated- $12,642,905
Total Profit	Total profit across all transactions	Calculated- $1,467,457
Profit Margin	Total profit divided by total sales, expressed as %	Calculated- 11.61%
Number of orders	Count of distinct order IDs	Calculated- 25,035
Number of Customers	Count of distinct customer IDs	Calculated- 4,873

Average Order Value	Total sales divided by number of distinct orders	Calculated- $505.01
Average Shipping Time	Average elapsed time between order date and ship date, in days	Calculated- 3.97 days

Expected deliverables

- Data Quality Assessment – data quality test cases, results, and findings.
- SQL Analysis – SQL queries for business analysis and data validation.
- Tableau Dashboard(s) – visualization of sales performance, profitability, and discount patterns.
- Business Recommendations – recommendations based on analysis findings.
- Executive Findings – a concise summary of key findings and business implications.

Scope

In Scope

- Historical sales and order analysis
- Customer and product performance analysis
- Regional and product-category performance analysis
- Profitability and discount analysis
- Shipping cost and shipping-duration analysis
- Data quality assessment and validation
- Tableau dashboard and business recommendations

Out of Scope

- Predicting future sales or building forecasting models
- Implementing changes to company pricing or discount policies
- Development of customer-level marketing campaigns
- Implementing actual operational change within the business

Data Profiling Summary

Dataset Overview

The dataset contains 51,290 transaction records across 27 fields covering sales, customers, products, geographic markets, order information, shipping, discounts and profitability from 2011-2014.

Initial Data Quality Findings

Findings	Records Count	Initial Assessment
Missing Values	0	No completeness issue identified in the selected required fields tested
Duplicate Row.ID values	0	All Row IDs were unique
Zero Sales	1	Requires business clarification to determine whether the zero-sales transaction is valid or represents a data issue
Negative profit	12,544	Required business interpretation; may represent valid business transactions
Market/Market2 mismatch	10,378	Requires clarification of geographic field definitions
Invalid shipping sequence	0	No date-sequence issue identified
Negative sales	0	No negative sales records identified